

Scalability & Future Plan

Date	2 July 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Lack of Database Persistence: Prediction queries and outcomes are processed in-memory and are not saved in a back-end database.	Admin users and loan officers cannot view transaction histories, track loan approval volumes, or audit model decisions over time.	High
2	Stateless Single-Instance Deployment (512MB RAM): The application runs on a single container instance on IBM Cloud with tight memory boundaries.	Under peak load (e.g., spike test scenarios), requests queue up, leading to high latency or potential Out-of-Memory (OOM) container crashes.	Medium
3	Static Offline Model Pipeline (rdf.pkl): The XGBoost model is trained offline in a Jupyter Notebook and manually serialized. It cannot adapt on its own.	The application cannot adapt automatically to changing economic conditions (e.g., changes in interest rates or shifting applicant demographics).	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Managed by a single Gunicorn container instance on IBM Code Engine. It is prone to performance degradation during concurrent request spikes.	Configure Horizontal Pod Auto-scaling (HPA) in Code Engine to automatically scale from 1 to 10 instances when CPU utilization crosses a 70% threshold.
2	Data Storage	Zero database persistence; client-side browser memory (localStorage) is used only for temporary draft saves.	Integrate a managed relational database service (e.g., IBM Cloud Databases for PostgreSQL or IBM Db2) to persist applicant profiles and model metadata.

3	Performance	Synchronous backend evaluation: every request executes standard scaling (scale.pkl) and inference classification (rdf.pkl) from disk.	Implement a Redis caching layer to cache predictive decisions for common input profiles, bypassing model computation for duplicate queries.
4	Security	Open endpoint; performs form boundary checks (parse_and_validate_form) but lacks client authentication or input encryption.	Add user authentication (via Flask-Login and JWT) to protect the /submit endpoint, and apply AES-256 field-level encryption for sensitive income data.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Relational Database Logs & Officer Portal	Month 3	Allows administrators to audit historical prediction runs and view approval rate analytics.
Phase 3	Continuous Integration & MLOps Pipeline	Month 6	Automatically triggers model retraining (using MLflow/Airflow) when new labeled loan datasets are updated, preventing model drift.
Phase 4	Explainable AI (XAI) Integration	Month 9	Integrates SHAP (SHapley Additive exPlanations) values to output visual feature-contribution charts, explaining exactly why a loan was rejected.